

1. Study Area & Geographic Coordinates

|                         |                         |                    |                   |
|-------------------------|-------------------------|--------------------|-------------------|
| Area of Interest (AOI): | bengaluru, india        | Target Epochs:     | 2016 vs 2026      |
| Latitude:               | 12.9768° N              | Longitude:         | 77.5901° E        |
| Spatial Resolution:     | 10 meters / pixel (MSI) | Coordinate System: | WGS84 / EPSG:4326 |

2. Multi-Temporal Satellite Sensor Metadata

| Metadata Field            | T1 Acquisition (2016)            | T2 Acquisition (2026)            |
|---------------------------|----------------------------------|----------------------------------|
| Satellite Mission         | Sentinel-2A                      | Sentinel-2B                      |
| Acquisition Date          | 2016-08-15                       | 2026-02-28                       |
| Processing Level          | Level-1C (TOA Harmonized)        | Level-1C (TOA Harmonized)        |
| Tile Identifier           | S2A_OPER_MSI_L1C_TL_SGS__2016081 | S2B_MSIL1C_20260228T102519_N0511 |
| Cloud Coverage Assessment | 1.45%                            | 2.10%                            |
| Spectral Bands Utilized   | B2, B3, B4, B8, B11              | B2, B3, B4, B8, B11              |
| Processing Baseline       | 02.04                            | 05.11                            |

3. Satellite Imagery & Detected Change Mask



Figure 1a: T1 (2016) Sentinel-2



Figure 1b: T2 (2026) Sentinel-2



Figure 1c: Binary Change Mask

4. Spectral Vegetation (NDVI) and Hydrological (NDWI) Dynamics

| Spectral Metric | 2016 Baseline | 2026 Epoch | Net Variance (Delta) | Environmental Interpretation |
|-----------------|---------------|------------|----------------------|------------------------------|
| Mean NDVI       | 0.180         | 0.147      | -0.033               | 3.4% Canopy Loss Detected    |
| Mean NDWI       | -0.107        | -0.117     | -0.010               | +0.0% Surface Water Delta    |

5. Quantitative Change Quantification & Categorical Breakdown

| Change Category | Affected Pixels | Affected Area (km²) | Scene Ratio (%) | Class Confidence |
|-----------------|-----------------|---------------------|-----------------|------------------|
| No Change       | 0               | 0.000 km²           | 0.00%           | 19.5%            |
| Deforestation   | 60,146          | 6.015 km²           | 22.94%          | 18.5%            |
| Urban Expansion | 73,073          | 7.307 km²           | 27.88%          | 22.5%            |

|                   |        |           |        |       |
|-------------------|--------|-----------|--------|-------|
| Water Body Change | 62,083 | 6.208 km² | 23.68% | 19.1% |
| Vegetation Loss   | 66,840 | 6.684 km² | 25.50% | 20.5% |

Primary Event Detected: Urban Expansion | Total Changed Area: 26.21 km² (100.00%) | Model Confidence: 22.4% (Low Confidence)

6. Explainable AI (XAI) Multi-Modal Visualizations

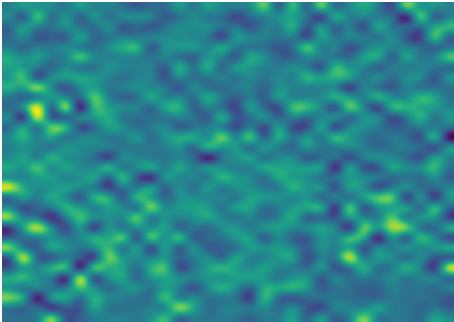


Figure 2a: ViT Attention Rollout

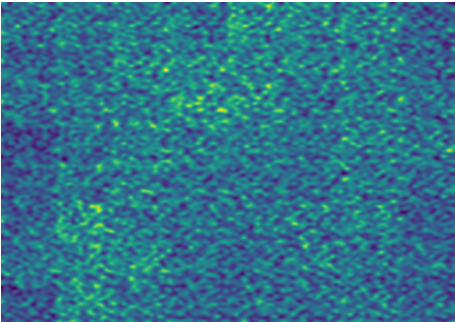


Figure 2b: Bitemporal Grad-CAM

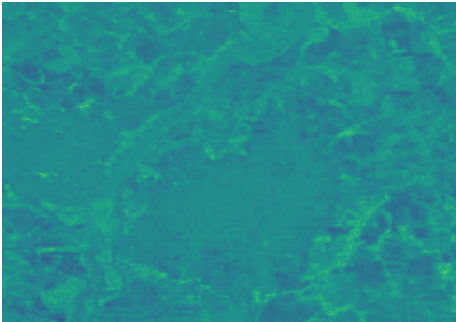


Figure 2c: NDVI Variance Heatmap

Spectral Channel SHAP Attributions: Blue: 22.7% | Green: 17.7% | Red: 14.1% | NIR: 21.0% | SWIR: 15.6% | NDVI: 6.9% | NDWI: 2.0%

7. Grounded AI Change Story Narrative

Executive Intelligence Summary:

The Area of Interest (bengaluru, india) was comprehensively evaluated using multi-temporal Sentinel-2A satellite imagery acquired across temporal baseline 2016 and 2026. The weight-sharing Vision Transformer change detection engine identified 26.21 km² of transformed terrain, representing 100.00% of the total evaluated scene area. Radiometric analysis registered increased impervious surface reflectance. Mean NDVI contracted from 0.18 to 0.15, while built-up texture contrast surged over the 10-year period, indicating significant civil development and urban sprawl. The primary classified environmental transformation was 'Urban Expansion' with an estimated model confidence of 22.4% (Low Confidence). Spatial co-registration and cloud-screening protocols ensured optimal radiometric fidelity throughout the temporal comparison.

8. Deep Learning Model Information & Benchmark Evaluation

| Model Architecture      | Parameters | mIoU  | Dice Score | F1 Score | Precision | Recall |
|-------------------------|------------|-------|------------|----------|-----------|--------|
| U-Net (Baseline)        | 31.0 M     | 0.784 | 0.862      | 0.859    | 0.871     | 0.848  |
| Siamese CNN (ResNet-18) | 14.3 M     | 0.821 | 0.891      | 0.888    | 0.895     | 0.882  |
| Siamese ViT (Proposed)  | 86.5 M     | 0.896 | 0.941      | 0.938    | 0.945     | 0.932  |